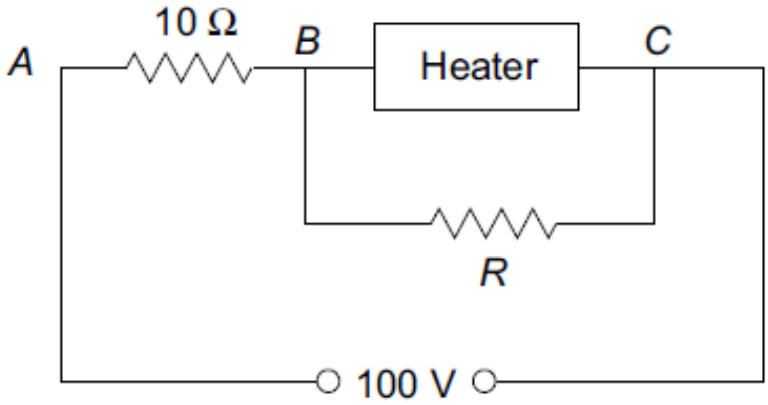
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A heater is designed to operate with a power of $1000\ \text{W}$ in a $100\ \text{V}$ line. It is connected, in combination with a resistance of 10Ω and a resistance R to a $100\ \text{V}$ line as shown in Fig. What should be the value of R so that the heater operates with a power of $62.5\ \text{W}$?



(A) 10Ω

(B) 62.5Ω

(C) $\frac{1}{5}\Omega$

(D) 5Ω

Q.2

A battery of emf E and internal resistance r is connected to a resistor of resistance r_1 and Q joules of heat is produced in a certain time t. When the same battery is connected to another resistor of resistance r_2, the same quantity of heat is produced in the same time t, the value of r is
(The heat produced due to the internal resistance of the battery can be neglated)

(A) $\frac{r_1^2}{r_2}$

(B) $\frac{r_2^2}{r_1}$

(C) $\frac{1}{2}(r_1 + r_2)$

(D) $\sqrt{r_1 r_2}$

Q.3

Charges on the capacitors in four oscillating LC circuits vary as follows :
(1) $q = 2\cos 4\text{t}$,
(2) $q = 4\cos \text{t}$,

@IITJEE_Advanced

(3) $q = 3\cos 4t$,
(4) $q = 4\cos 2t$,
with q in coulomb and t in second. In which circuit(s) current amplitude is greatest?

(A) (1)

(B) (2)

(C) (3)

(D) (4)

Q.4

Drift velocity $\mathrm{V_d}$ varies with the intensity of electric field as per the relation

(A) $\mathrm{V_d} \propto \mathrm{E}$

(B) $\mathrm{V_d} \propto \frac{1}{\mathrm{E}}$

(C) $\mathrm{V_d} =$ constant

(D) $\mathrm{V_d} \propto \mathrm{E}^2$

Q.5

A conducting wire of cross-sectional area $1\ \mathrm{cm}^2$ has 3×10^{23} charge carriers per m^3. If wire carries a current of 24 mA, then drift velocity (in $\mathrm{m/sec}$) of carriers is

(A) 0.5

(B) 0.05

(C) 0.0005

(D) 0.005

Q.6

The total momentum of electrons in a straight wire of length $1000\ \mathrm{m}$ carrying a current of $70\ \mathrm{A}$ is closest to

(A) $40 \times 10^{-8}\ \mathrm{N-sec}$

(B) $30 \times 10^{-8}\ \mathrm{N-sec}$

(C) $50 \times 10^{-8}\ \mathrm{N-sec}$

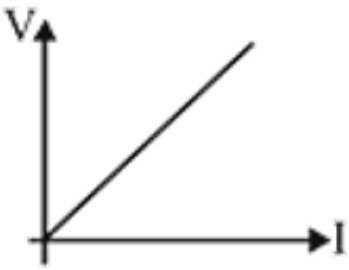
(D) $70 \times 10^{-8}\ \mathrm{N-sec}$

Q.7

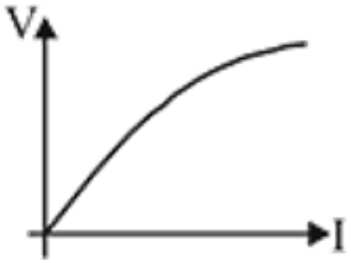
Suppose the drift velocity v_d in a material varied with the applied electric field E as $\mathrm{v_d} \propto \sqrt{\mathrm{E}}$ Then V - I graph for a wire made of such a material is best given by :

@IITJEE_Advanced

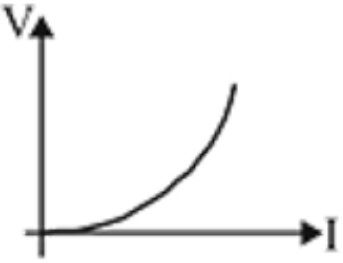
(A)



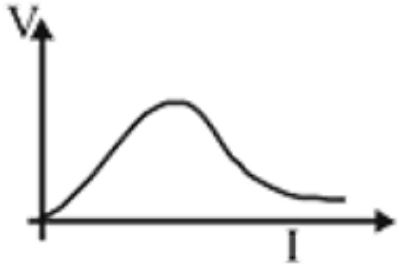
(B)



(C)



(D)



Q.8

The current density varies with radial distance as $J = ar^2$, in a cylindrical wire of radius R. The current passing through the wire between radial distance $R/3$ and $R/2$ is

(A) $\frac{65\pi aR^4}{2592}$

(B) $\frac{25\pi aR^4}{72}$

(C) $\frac{65\pi a^2R^3}{2938}$

(D) $\frac{81\pi a^2R^4}{144}$

Q.9

In a certain particle accelerator, electrons emerge in pulses at the rate of 250 pulses per second. Each pulse is of duration of $200\ \text{ns}$ and the electrons in the pulse constitute a current of $250\ \text{mA}$. The number of electrons delivered by the accelerator per pulse is

(A) 8.00×10^{10}

@IITJEE_Advanced

(B) 5.00×10^{8}

(C) 3.13×10^{11}

(D) 9.60×10^{10}

Q.10

Drift velocity of electrons is due to

(A) motion of conduction electrons due to random collisions.

(B) motion of conduction electrons due to electric field $\vec{E}$

(C) repulsion to the conduction electrons due to inner electrons of ions.

(D) collision of conduction electrons with each other.

Q.11

A metal wire is subjected to a constant potential difference. When the temperature of the metal wire increases, the drift velocity of the electron in it

(A) increases, thermal velocity of the electron increases

(B) decreases, thermal velocity of the electron increases

(C) increases, thermal velocity of the electron decreases

(D) decreases, thermal velocity of the electron decreases

Q.12

The amount of charge Q passed in time t through
a cross-section of a wire is $\mathrm{Q} = 5\mathrm{t}^2 + 3\mathrm{t} + 1$. The value of current at time $\mathrm{t} = 5\ \mathrm{s}$ is

(A) $9\ \mathrm{A}$

(B) $49\ \mathrm{A}$

(C) $53\ \mathrm{A}$

(D) None of these

Q.13

A conductor carries a current of $50\mu\mathrm{A}$. If the area of cross-section of the conductor is $50\ \mathrm{mm}^2$ then value of the current density in Am^{-2} is

(A) 0.5

(B) 1

(C) 10^{-3}

@IITJEE_Advanced

(D) 10^{-6}

Q.14

At room temperature, copper has free electron density of 8.4×10^{28} per m^3. The copper conductor has a cross-section of $10^{-6}\ \mathrm{m}^2$ and carries a current of $5.4\ \mathrm{A}$. The electron drift velocity in copper is

(A) $400\ \mathrm{m/s}$

(B) $0.4\ \mathrm{m/s}$

(C) $0.4\ \mathrm{mm/s}$

(D) $72\ \mathrm{m/s}$

Q.15

The number of free electrons per $100\ \mathrm{m}^2$ of ordinary copper wire is 2×10^{21}. Average drift speed of electrons is $0.25\ \mathrm{mm/s}$. The current flowing is

(A) $5\ \mathrm{A}$

(B) $80\ \mathrm{A}$

(C) $8\ \mathrm{A}$

(D) $0.8\ \mathrm{A}$

Answers & Solutions

Q.1 Answer:

5Ω

Solution:

Resistance of heater

$(R_{\mathrm{H}}) = \frac{V^2}{P} = \frac{100 \times 100}{1000} = 10\Omega.$

Current in the circuit is

$I = \frac{100}{10 + \frac{10R}{(10+R)}}$

$\therefore$ Current in heater is

$$I_{\mathrm{H}} = I \times \frac{R}{(10+R)} = \frac{100}{\left[10 + \frac{10R}{(10+R)}\right]} \times \frac{R}{(10+R)}$$

$$= \frac{10R}{\left[1 + \frac{R}{(10+R)}\right](10+R)} = \frac{5R}{(5+R)}$$

$\therefore \quad Q = I_{\mathrm{H}}^2 \times 10$

or $\frac{25R^2}{(5+R)^2} \times 10 = 62.5$ (given)

which gives $R = 5\Omega$. Hence the correct choice is (d).

@IITJEE_Advanced

Q.2 Answer:

$\sqrt{r_1 r_2}$

Solution:

In the first case, the current in the circuit is

$$I_1 = \frac{E}{r_1 + r}$$

$$\therefore \quad Q_1 = I^2 r_1 t = \left(\frac{E}{r_1 + r}\right)^2 \times r_1 t \quad \text{(i)}$$

In the second case,

$Q_2 = \left(\frac{E}{r_2 + r}\right)^2 \times r_2 t$ **(2)**

Equating (i) and (ii) we get

$$\frac{r_1}{(r_1 + r)^2} = \frac{r_2}{(r_2 + r)^2}$$

$$\text{Or } r_1 (r_2 + r)^2 = r_2 (r_1 + r)^2$$

Or $r_1 \left(r_2^2 + 2 r r_2 + r^2\right) = r_2 \left(r_1^2 + 2 r r_1 + r^2\right)$

Or $r^2 (r_1 - r_2) = r_1 r_2 (r_1 - r_2)$

or $r = \sqrt{r_1 r_2}$

Hence the correct choice is (d).

Q.3 Answer:

(3)

Solution:

$i_1 = \frac{dq}{dt} = \frac{d}{dt}(2\cos 4t) = -8\sin 4t$

$i_2 = \frac{dq}{dt} = \frac{d}{dt}(4\cos t) = -4\sin t$

$i_3 = \frac{dq}{dt} = \frac{d}{dt}(3\cos 4t) = -12\sin 4t$

$i_4 = \frac{dq}{dt} = \frac{d(4\cos 2t)}{dt} = -8\sin 2t$

Clearly, amplitude of i_3 **is greatest.**

Q.4 Answer:

$\mathrm{V_d} \propto \mathrm{E}$

Solution:

$\mathrm{V_d} = \frac{\mathrm{e}}{\mathrm{m}} \times \frac{\mathrm{v}}{\ell}\tau$ **or,** $\mathrm{V_d} = \frac{\mathrm{e}}{\mathrm{m}} \cdot \frac{\mathrm{E}\ell}{\ell}\tau [\because \mathrm{V} = \mathrm{E}\ell]$

$\therefore \mathrm{V_d} \propto \mathrm{E}$

Q.5 Answer:

0.005

Solution:

$\mathrm{I} = \mathrm{neAV_d}$

$\mathrm{V_d} = \frac{\mathrm{I}}{\mathrm{neA}} = 5 \times 10^{-3}\ \mathrm{m/sec}$

@IITJEE_Advanced

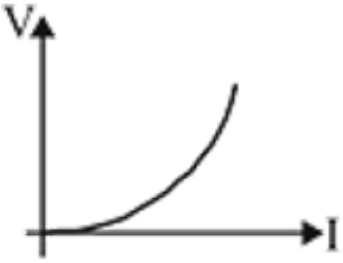
Q.6 Answer:

40×10^{-8} N − sec

Solution:

No. of electron in the wire $= nA\ell \quad (n = e^-$ **density) and momentum** $(nA\ell)m_e v_d = neAv_d \frac{\ell m_e}{e} = \frac{I\ell m_e}{e}$
$= \frac{70\times1000}{1.6\times10^{-19}} \times 9.1 \times 10^{-31} = 40 \times 10^{-8}$ N − sec

Q.7 Answer:



Solution:

$i = neAV_d$ **and** $V_d \propto \sqrt{E}$ **(Given)**
or, $i \propto \sqrt{E}$
$i^2 \propto E \quad i^2 \propto V$
Hence graph (c) correctly depicts the V **-Igraph for a wire made of such type of material.**

Q.8 Answer:

$\frac{65\pi aR^4}{2592}$

Solution:

Given ; $J = ar^2 . i = \int_1^2 J \times 2\pi r dr = \int_{R/3}^{R/2} ar^2 \times 2\pi r dr$
$= 2\pi a \int_{R/3}^{R/2} r^3 dr = 2\pi a \left| \frac{r^4}{4} \right|_{R/3}^{R/2}$
$= \frac{\pi a}{2} \left[\left(\frac{R}{2}\right)^4 - \left(\frac{R}{3}\right)^4 \right]$
$= \frac{\pi a R^4}{2} \times \frac{65}{81\times16} = \frac{65\pi aR^4}{2592}$

Q.9 Answer:

3.13×10^{11}

Solution:

The charge in a pulse $dq = Idt = 5.00 \times 10^{-8}$C.
$n = \frac{q}{e} = \frac{5\times10^{-8}C}{16\times10^{-19}C} = 3.12 \times 10^{11}$

Q.10 Answer:

motion of conduction electrons due to electric field $\vec{E}$

Solution:

@IITJEE_Advanced

Motion of conduction electrons due to random collisions has no preffered direction and average to zero. Drift velocity is caused due to motion of conduction electrons due to applied electric field $\vec{E}$.

Q.11 Answer:

decreases, thermal velocity of the electron increases

Solution:

When the temperature increases, resistance increases. As the e.m.f. applied is the same, the current density decreases the drift velocity decreases. But the rms velocity of the electron due to thermal motion is proportional to $\sqrt{T}$. Therefore, the thermal velocity increases.

Q.12 Answer:

$53\ \mathrm{A}$

Solution:

$\mathrm{I}=\frac{\mathrm{dQ}}{\mathrm{dt}}=10\mathrm{t}+3$
$\mathrm{Att}=5\ \mathrm{s}, \mathrm{I}=10\times5+3=53\ \mathrm{A}$

Q.13 Answer:

1

Solution:

$$\text{Current density } \mathrm{J}=\mathrm{I}/\mathrm{A}$$
$$=50\times16^{-6}/50\times10^{-6}=1\mathrm{Am}^{-2}$$

Q.14 Answer:

$0.4\ \mathrm{mm/s}$

Solution:

$\mathrm{v_d}=\frac{\mathrm{I}}{\mathrm{neA}}$ **Here,** $\mathrm{I}=5.4\ \mathrm{A}, \mathrm{n}=8.4\times10^{28}$, **per** m^3
$\mathrm{A}=10^{-6}\ \mathrm{m}^2, \mathrm{e}=1.6\times10^{-19}\mathrm{C}$
$\therefore \mathrm{v_d}=\frac{5.4}{84\times10^{28}\times16\times10^{-19}\times10^{-6}}=0.4\ \mathrm{mm/s}$

Q.15 Answer:

$8\ \mathrm{A}$

Solution:

$\mathrm{I}=\mathrm{neAv_d}$
$=2\times10^{21}\times1.6\times10^{-19}\times100\times0.25\times10^{-3}$
$=2\times16\times0.25=8=8\ \mathrm{A}$

@IITJEE_Advanced